

Second Terminal Examination - 2082

Class: XI Subject: Mathematics (0071) Full Marks: 75
Time: 3 hrs. SET 'A' Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- The number of non-empty subsets of the set $A = \{1, 2, 3, 4\}$ is
a) 15 b) 17 c) 14 d) 16
- If $A = [0, 5]$ and $B = (3, 7]$, then $A \cup B$ is
a) $(0, 7)$ b) $[0, 7)$
c) $[0, 7]$ d) $(0, 7]$
- The range of the real valued function $f(x) = \frac{|x|}{x}$ is
a) $\{-2, 2\}$ b) $\{-1, 1\}$
c) $[-1, 1]$ d) $\mathbb{R} - \{0\}$
- An exponential function $y = a^x$, becomes a real valued function if
a) $a > 0$ b) $a \neq 1$
c) $a < 0$ d) both (a) and (b)
- If α and β be two roots of the equation $x^2 + 7x + 10 = 0$ then the value of $\alpha^2 + \beta^2$ is
a) 29 b) 39 c) 19 d) 25
- $\sqrt{-25} \cdot \sqrt{-16} =$
a) 20 b) $20i$ c) -20 d) $-20i$

7. The distance of the point $(2, 3)$ from the line $3x + 4y - 3 = 0$ is
 a) 4 units b) 3 units c) 5 units d) 6 units
8. The lines represented by $ax^2 + 2hxy + by^2 = 0$ will be coincident if
 a) $h^2 = 2ab$ b) $a + b = 0$
 c) $a - b = 0$ d) $h^2 = ab$
9. $\lim_{x \rightarrow 2^+} \frac{|x-2|}{x-2} =$
 a) -1 b) 1 c) 0 d) does not exist
10. The derivative of $\cos x$ with respect to $\sin x$ is
 a) $\tan x$ b) $\cot x$ c) $-\cot x$ d) $-\tan x$
11. The derivative of $\tan^{-1} x$ with respect to x is
 a) $\frac{1}{1-x^2}$ b) $\frac{-1}{1+x^2}$ c) $\frac{1}{1+x^2}$ d) $\frac{1}{\sqrt{1-x^2}}$

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) Construct the truth table of $q \vee (p \vee \sim q)$. Also, comment on the result. [1+2]
 b) Solve the inequality: $|2x - 1| \leq 3$. [2]
13. a) Let the function $f: A \rightarrow B$ be defined by $f(x) = \frac{x+1}{2x-1}$ with $A = \{-1, 0, 1, 2, 3, 4\}$ and $B = \{-1, 0, \frac{4}{5}, \frac{5}{7}, 1, 2, 3\}$. Find the range of f . Is the function f one to one and onto? If not, how can the function be made one to one and onto both? [1+1+1]
 b) If $a^2 + b^2 = 7ab$, then prove that : $\log\left(\frac{a+b}{3}\right) = \frac{1}{2}(\log a + \log b)$. [2]
14. a) Find the domain and range of the real valued function
 $f(x) = \sqrt{21 - 4x - x^2}$. [3]
 b) Find the value of x and y if $(x + 2) + yi = (3 + i)(1 - 2i)$. [2]

15. a) Find all possible values of z if $z^2 = 5 + 12i$. [3]
 b) If α and β are the roots of the equation $px^2 + qx + q = 0$, then show that: $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{q}{p}} = 0$. [2]
16. a) Let the equations of two straight lines be $3x + 4y - 5 = 0$ and $4x + 3y + 7 = 0$.
 i) Find the angle bisectors between them. [1]
 ii) Show that angle bisectors are perpendicular. [1]
 iii) Also, identify the bisector which contains origin. [1]
 b) If p be the length of perpendicular from origin on the line $\frac{x}{a} + \frac{y}{b} = 1$, then prove : $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$. [2]

17. a) Evaluate : $\lim_{x \rightarrow \theta} \frac{x \sin \theta - \theta \sin x}{x - \theta}$ [3]
 b) If the equation $(1 + n^2)x^2 + 2ndx + d^2 - a^2 = 0$ has equal roots, show that : $d^2 = a^2(1 + n^2)$. [2]
18. a) From the first principle, find the derivative of $\sqrt{2x + 3}$. [3]
 b) If $x = a(t + \sin t)$ and $y = a(1 - \cos t)$ then prove that $\frac{dy}{dx} = \tan \frac{t}{2}$. [2]
19. a) Write the derivative of $\sec x$ and $\cot x$ w.r.t. x . [1]
 b) Find $\frac{dy}{dx}$ if $x^2 + y^2 = \sin(xy)$. [2]
 c) Evaluate : $\lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x} - \sqrt{x - a})$ [2]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Define $A - B$ in set builder form. If A , B and C be three non-empty sets, then prove that : $A - (B \cup C) = (A - B) \cap (A - C)$ [1+2]
 b) Let $f: \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{1\}$ be given by $f(x) = \frac{x}{x-2}$. Show that f is bijective function. [1+2]
 c) Solve the inequality: $|2x - 5| \leq 7$ and show the solution set in graph. [1+1]

21. a) Write the condition that the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ may have a root common. Find the values of k for which the equations $x^2 - kx - 21 = 0$ and $x^2 - 3kx + 35 = 0$ may have a common root. [1+2]
- b) Write the condition for a general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines. Find the value 'k' for which the equation $x^2 + kxy + y^2 - 5x - 7y + 6 = 0$ represents a pair of lines. [1+2]
- c) If the distance between two parallel lines $2x + 3y - 7 = 0$ and $4x + 6y + k = 0$ is $\frac{23}{2\sqrt{13}}$ units, then find value(s) of k . [2]
22. a) Define continuity of a function $f(x)$ at any point $x = a$. [1]
- b) Let a function $f(x)$ be defined by
- $$f(x) = \begin{cases} 2 - x^2 & \text{for } x < 2 \\ 3 & \text{for } x = 2 \\ x - 4 & \text{for } x > 2 \end{cases}$$
- i) Verify that the limit of function exists. [1]
- ii) Is function continuous at $x = 2$? Give reason. [1]
- iii) If $f(x)$ is not continuous, mention the type of discontinuity. [1]
- iv) Redefine the function to make it continuous at $x = 2$. [1]
- c) Find $\frac{dy}{dx}$ if $x^3 + y^3 - 3axy = 0$. [3]

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